

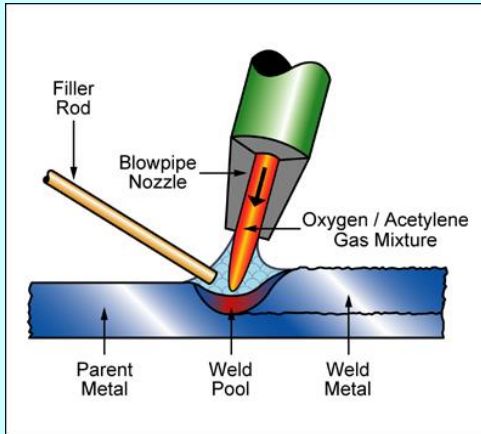
Lecture 11 - Using Adhesives

in Assembly and Modern Manufacture

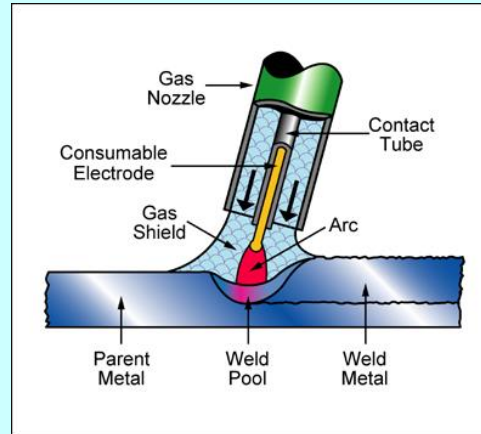
James Busfield



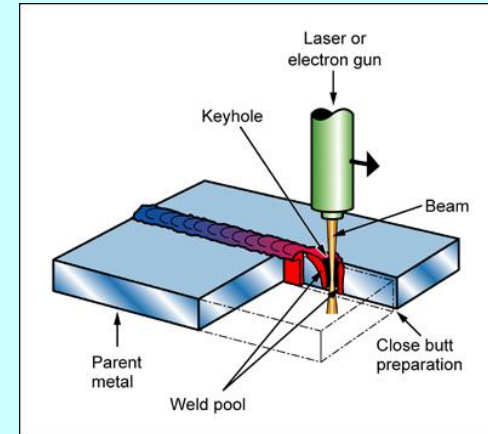
Oxyacetylene



Metal Inert Gas

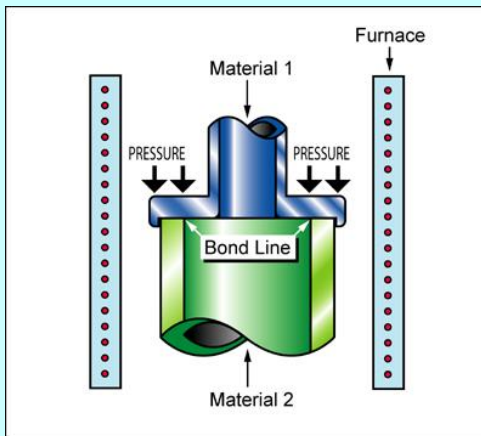


Electron Beam



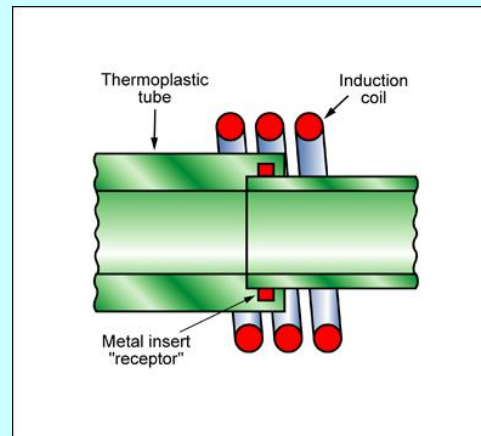
Metal Welding

Diffusion Bonding



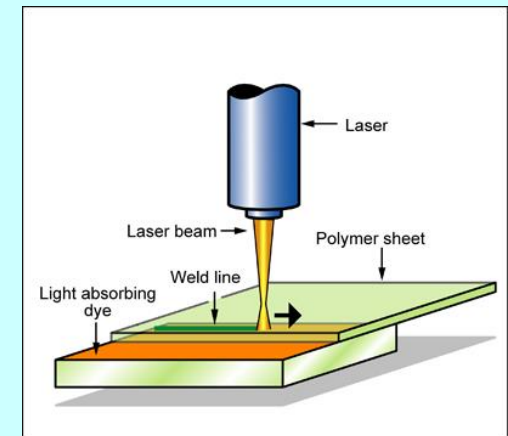
Ceramic Welding

Induction Welding



Polymer Welding

Laser Welding



Increasing use of composites which require good bonding to work



1980s

A310
B767

5-6%



2000s

A380

23%



1990s

A320 B777

10-15%

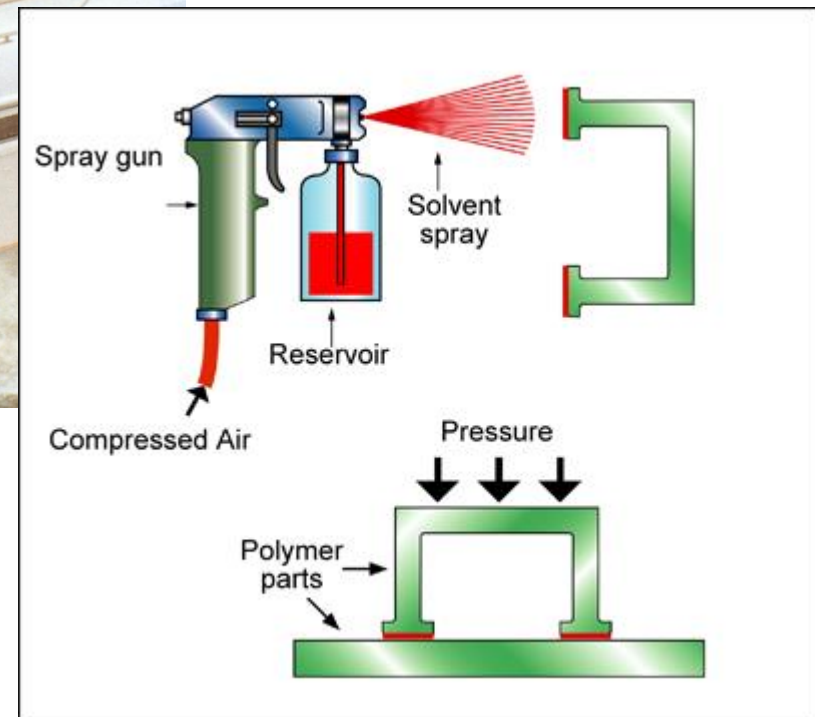


2020s

A350 B787

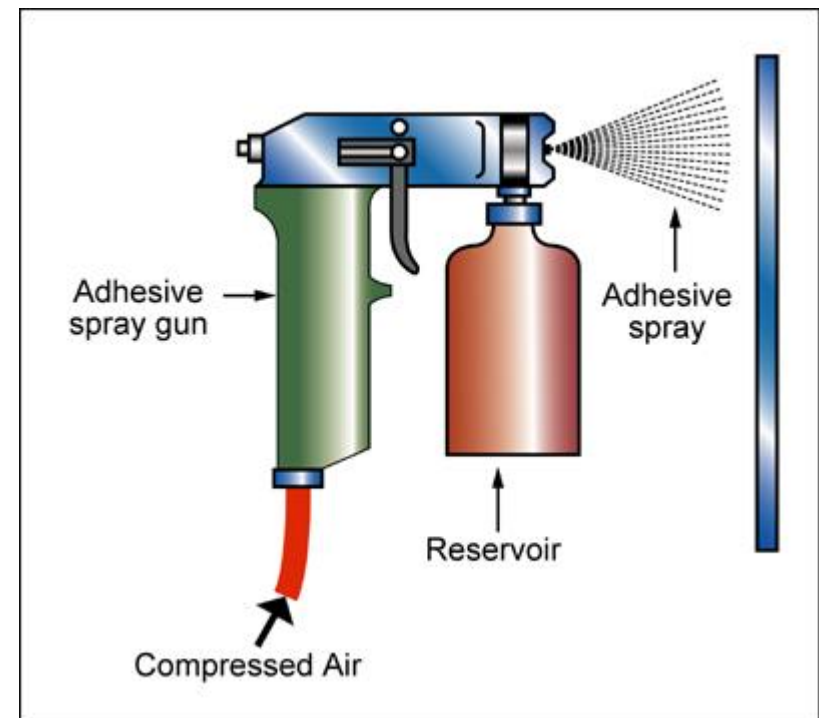
50%+

Modern Luxury Yacht



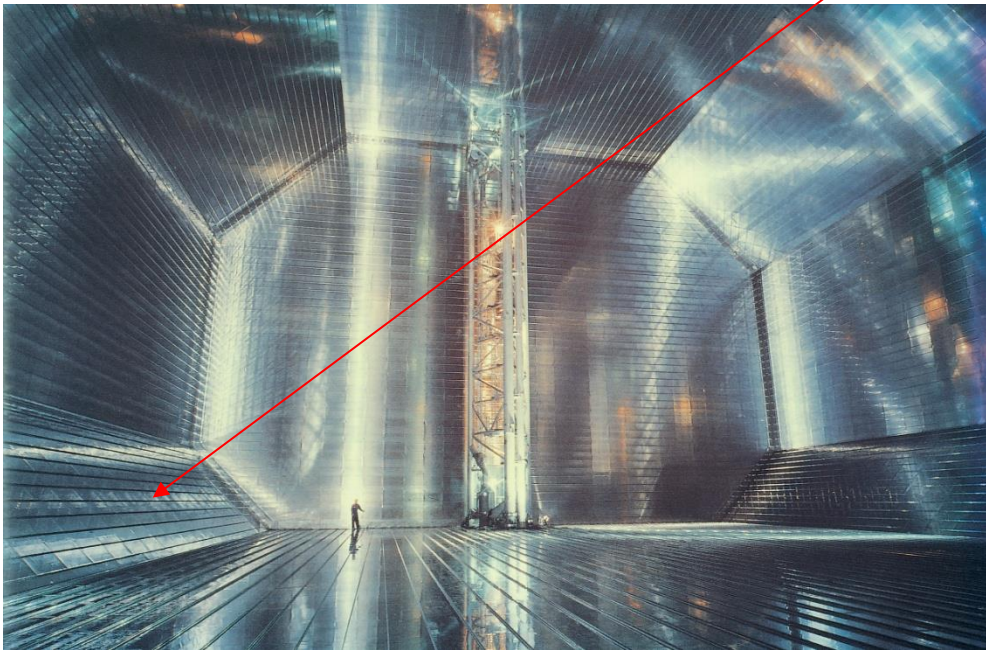
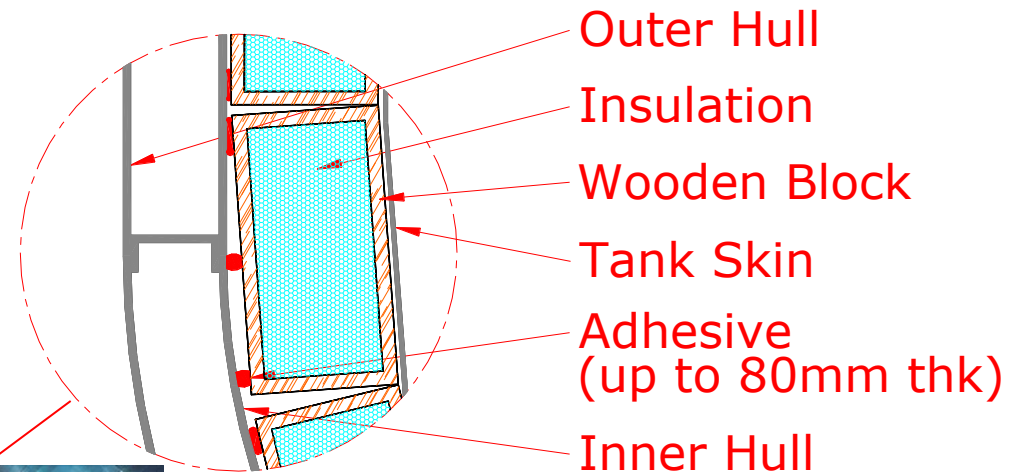
Adhesive Bonding

- Attaching bows to slippers
- Instant Adhesive replaced stitching



LNG gas tanker

- Gas Tanker
- Bonding and Sealing of internal panels



Types of Adhesive

Polyurethane

Epoxy

MS Polymer

Polysulphide

PVA

Cyanoacrylate

UV Cure

Anaerobic

Hot melt

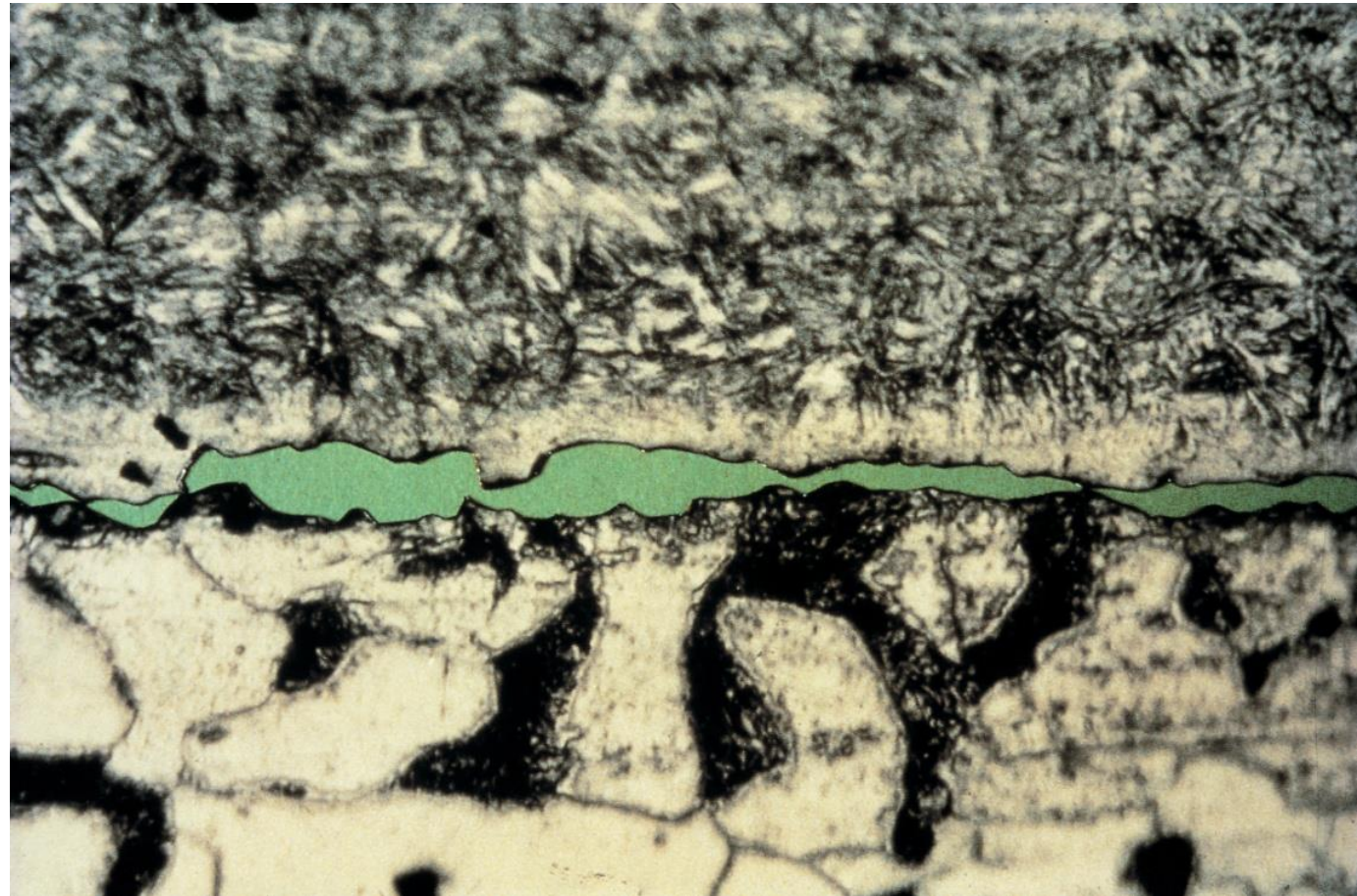
Toughened Acrylic

Phenolic

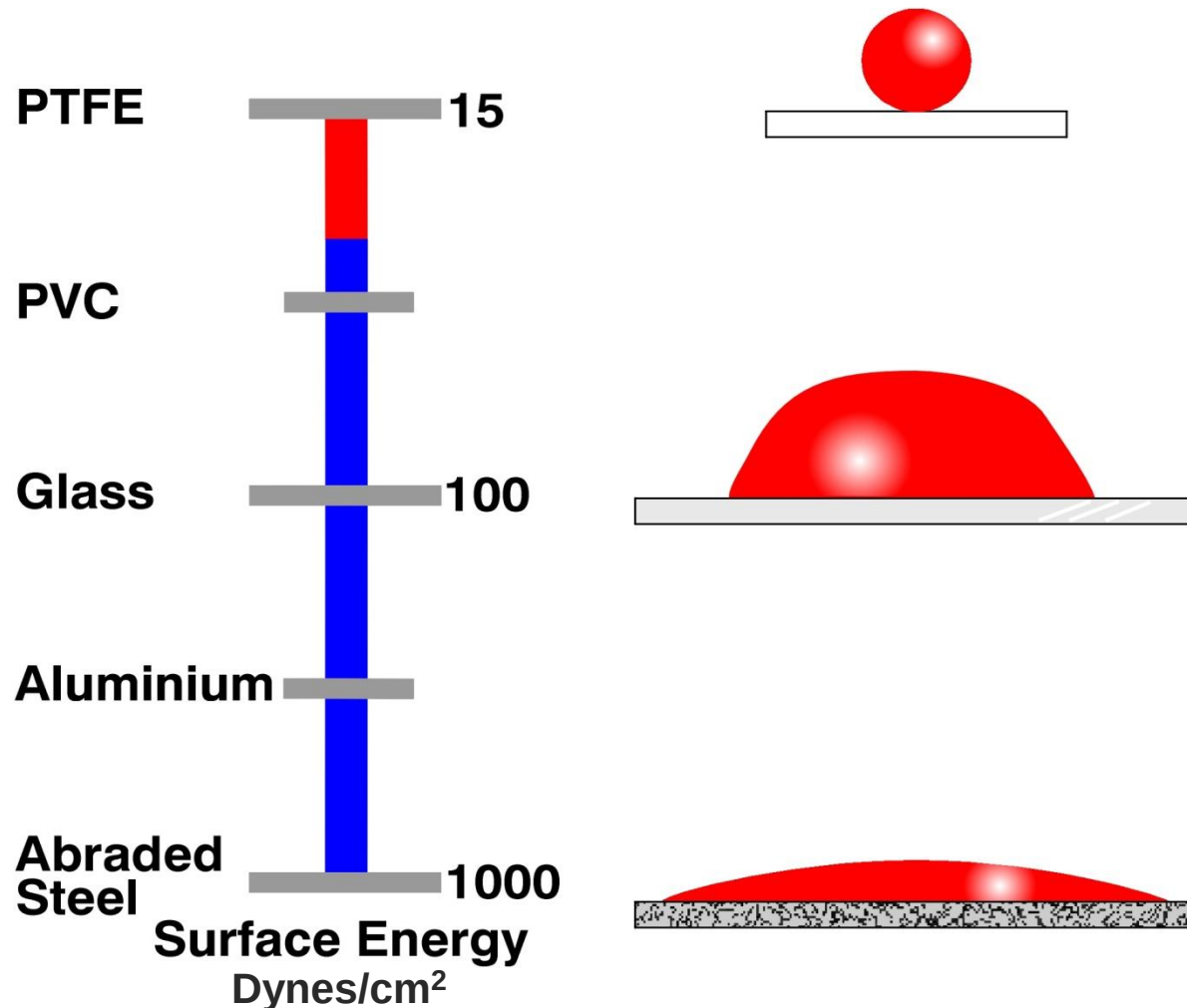
Silicone

Surface Irregularity

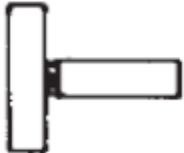
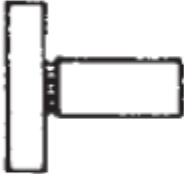
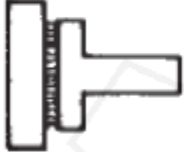
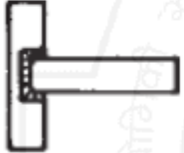
- Micrograph showing surface irregularity
- Adhesive fully fills the joint



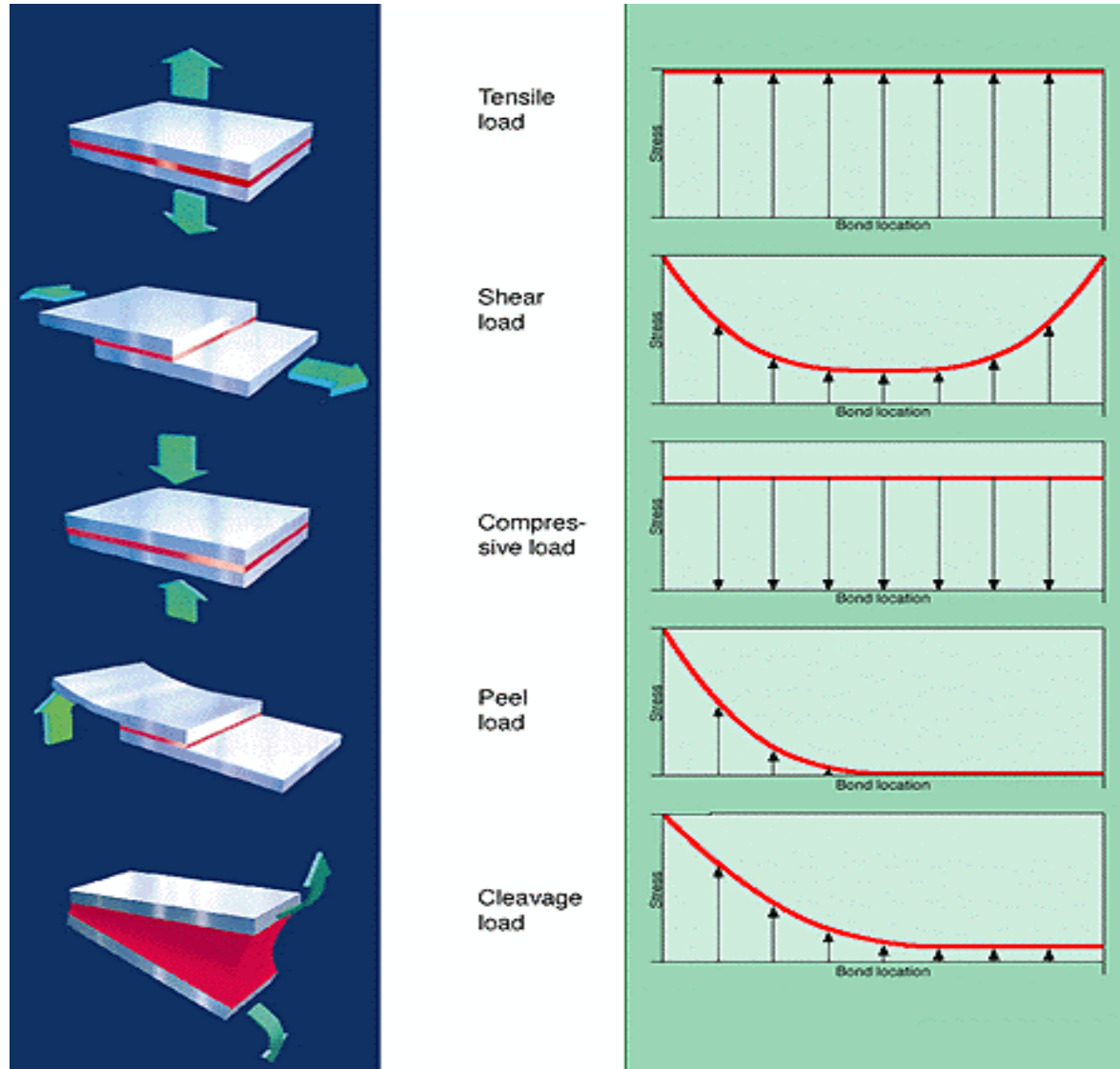
Surface wetting



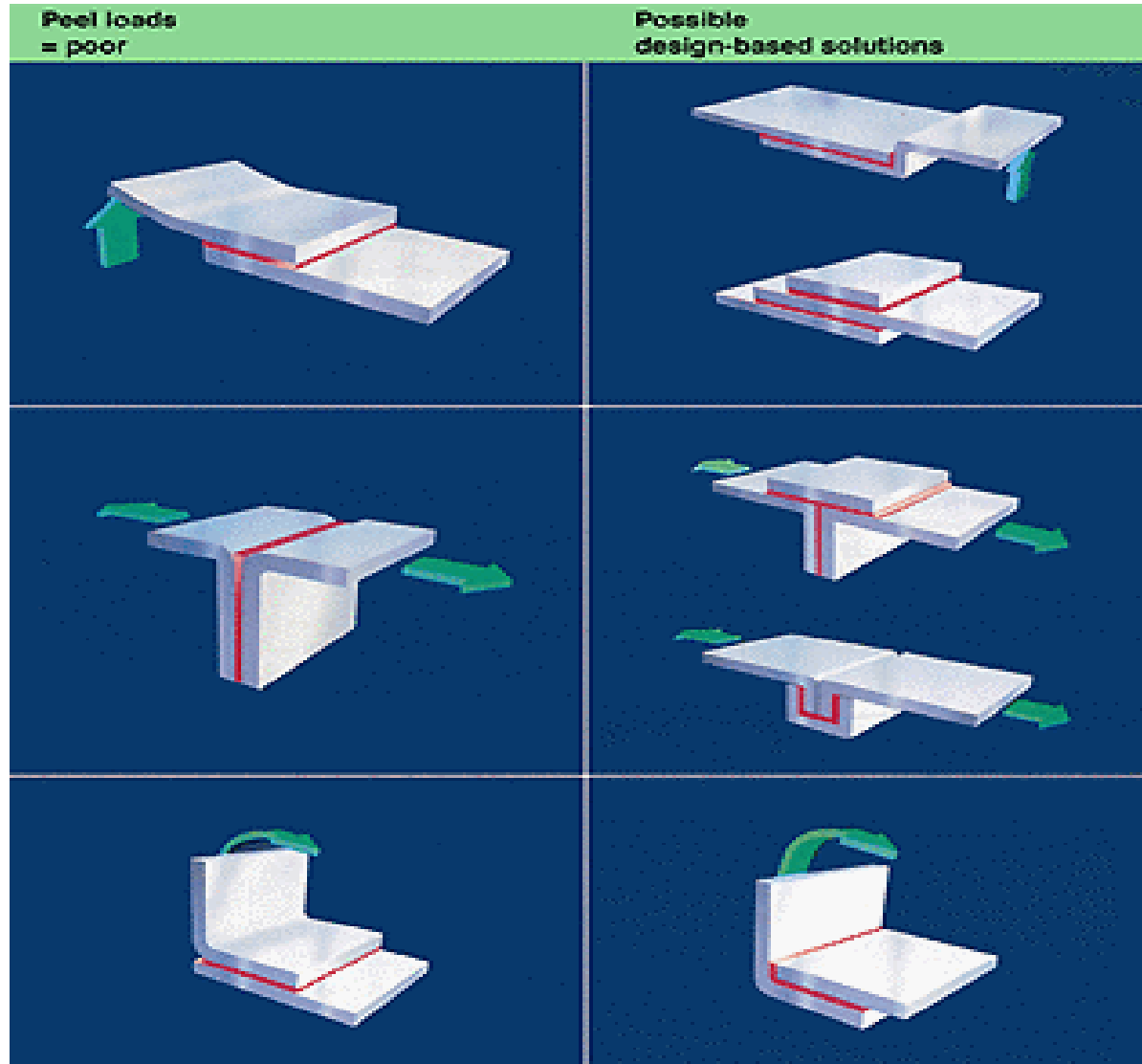
Butt joint design

Modified butt joint design	Performance
	Poor
	Better in tension, poor in bending
	Poor
	Better
	Better yet
	Good in tension and bending, costs more
	Excellent but expensive, slower

Avoid / reduce stress peaks



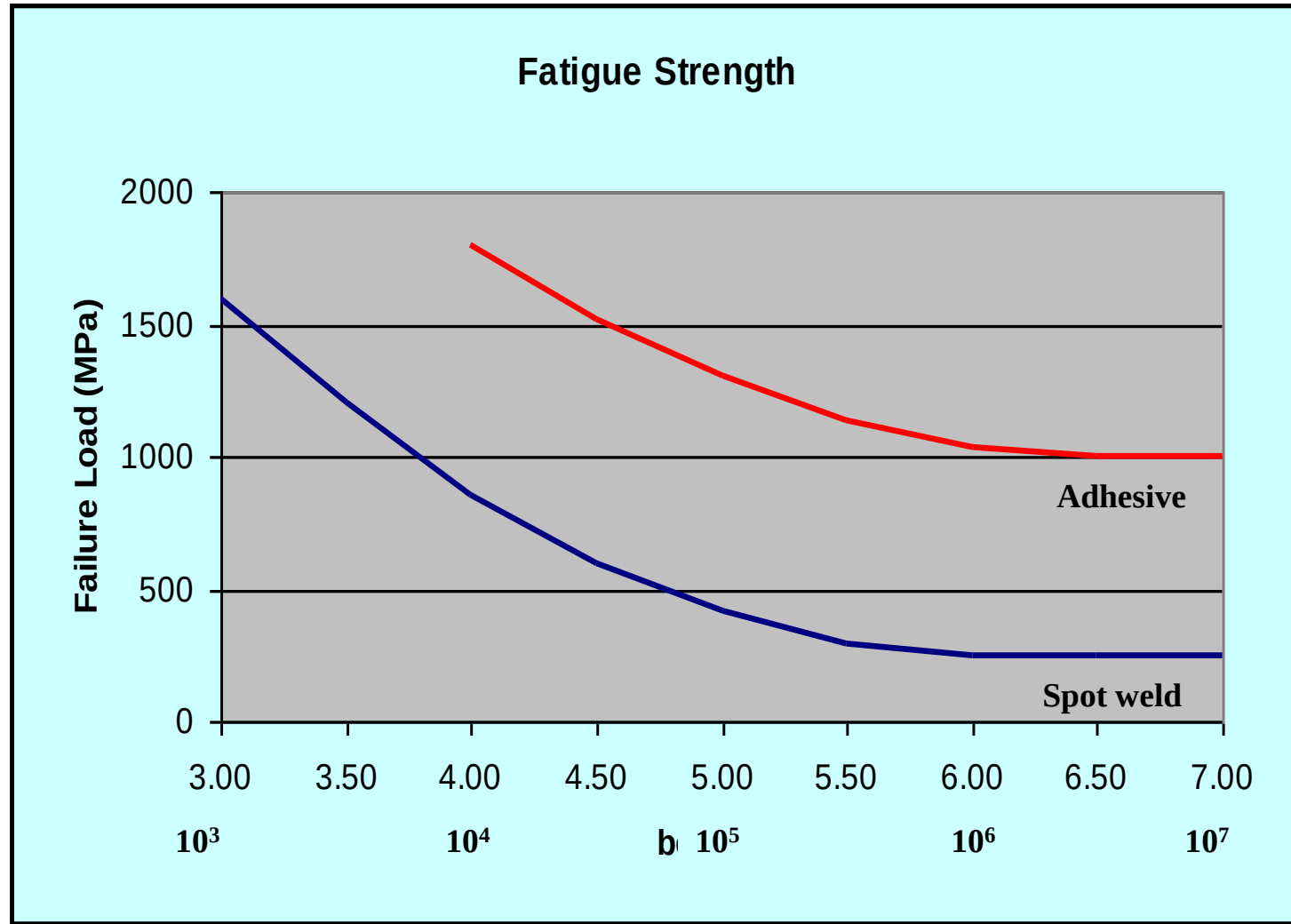
Avoid or at least reduce stress peaks



- Adhesive is the only option to join the two materials



Fatigue Strength

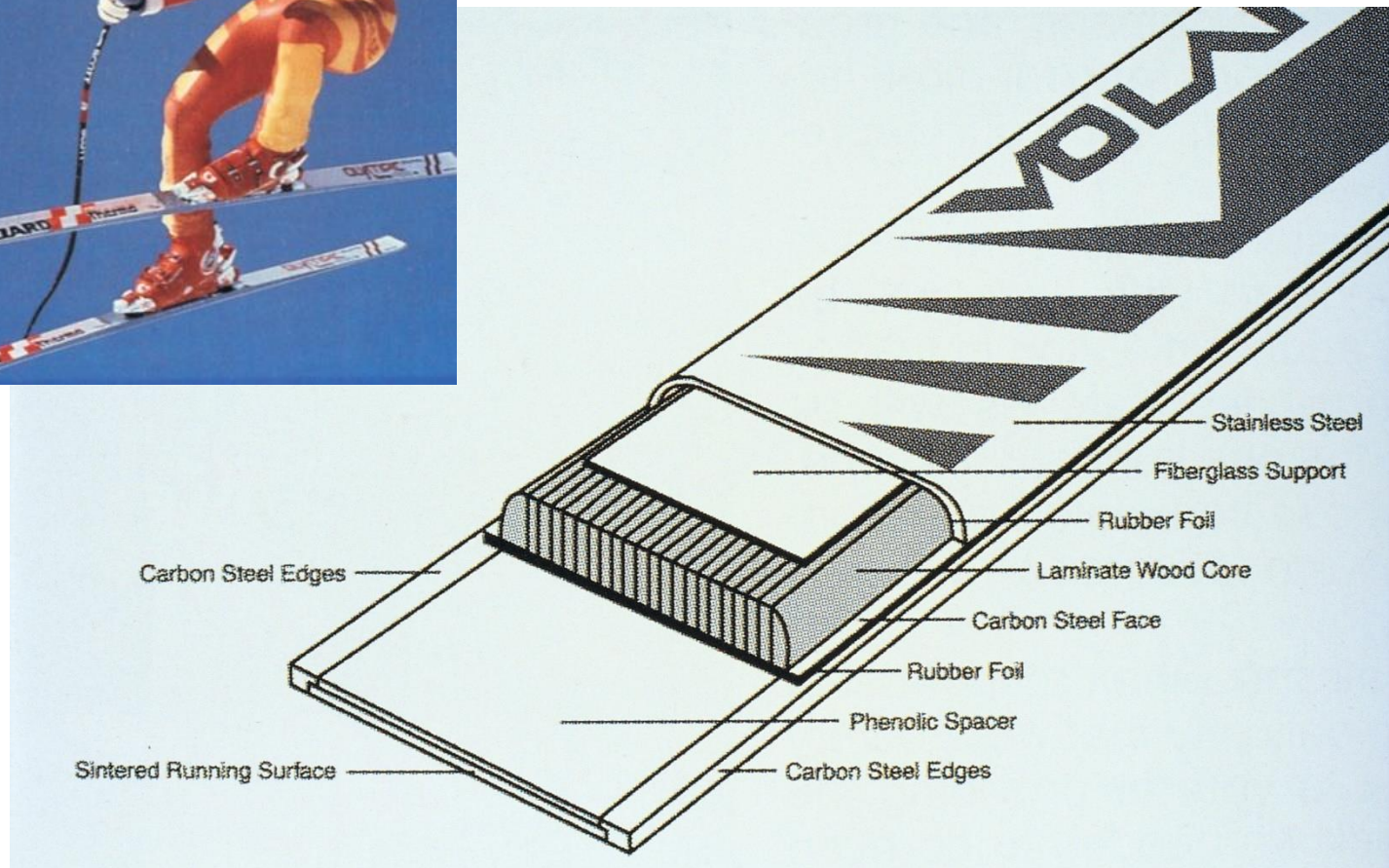


Adhesive & Cylindrical Bonding

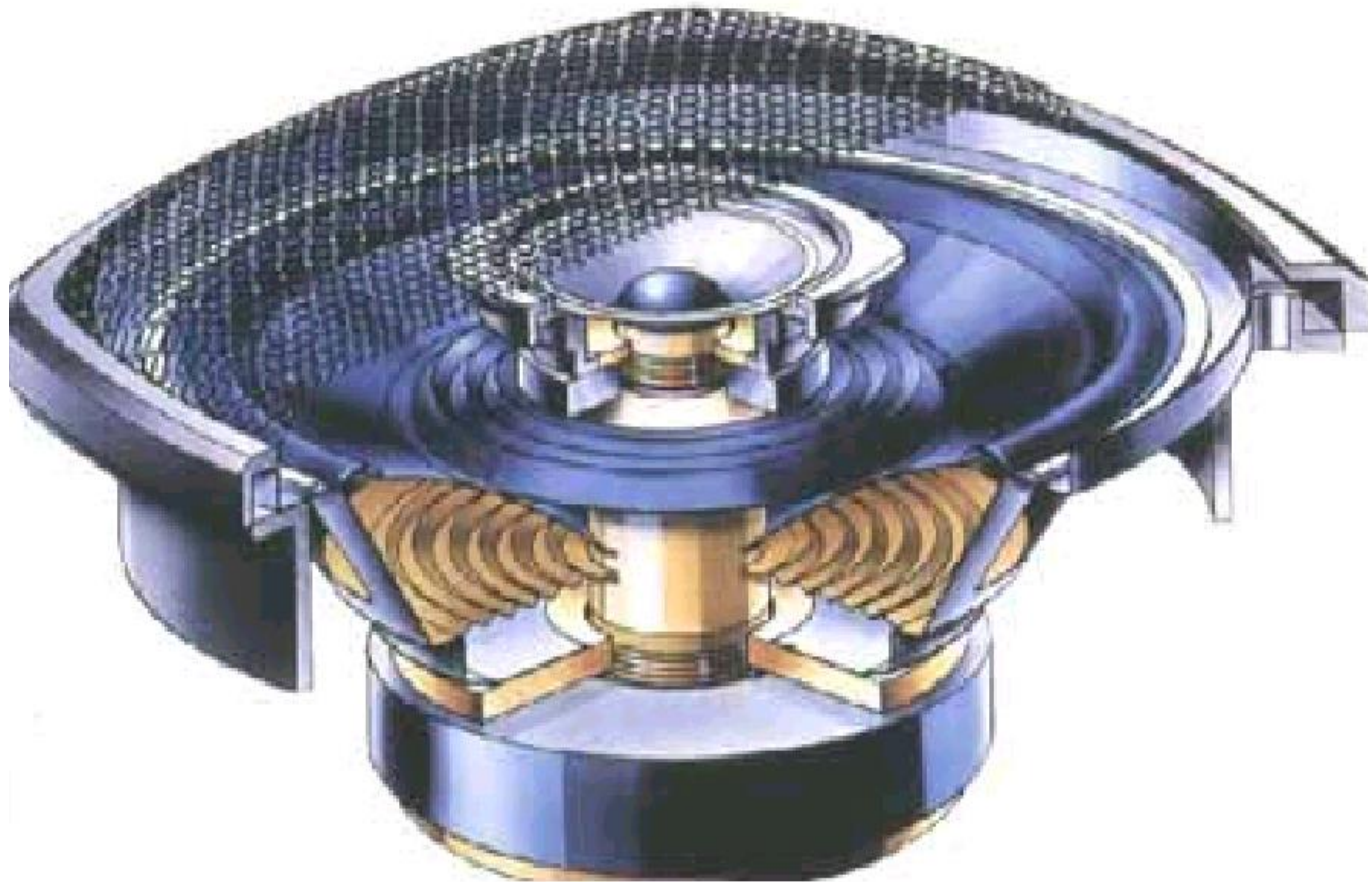
- Fatigue Resistance



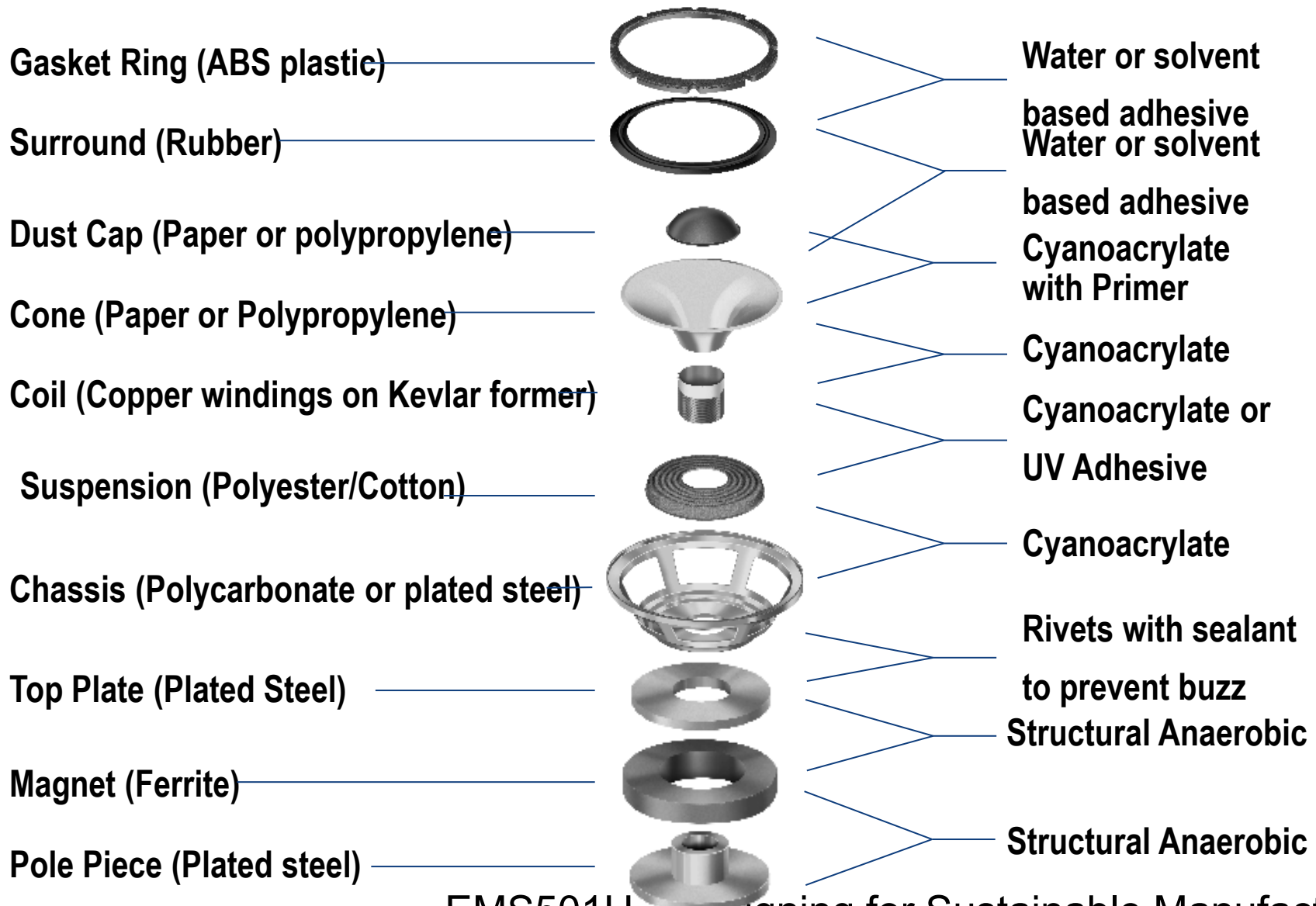
SKI design



Loud Speaker Design



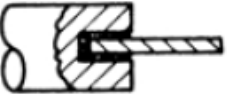
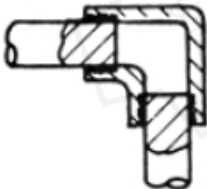
Loudspeaker Assembly



Composite Bonding

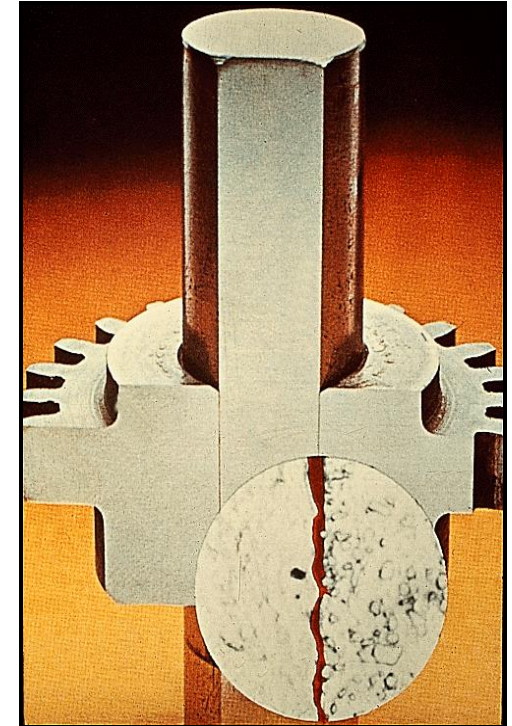
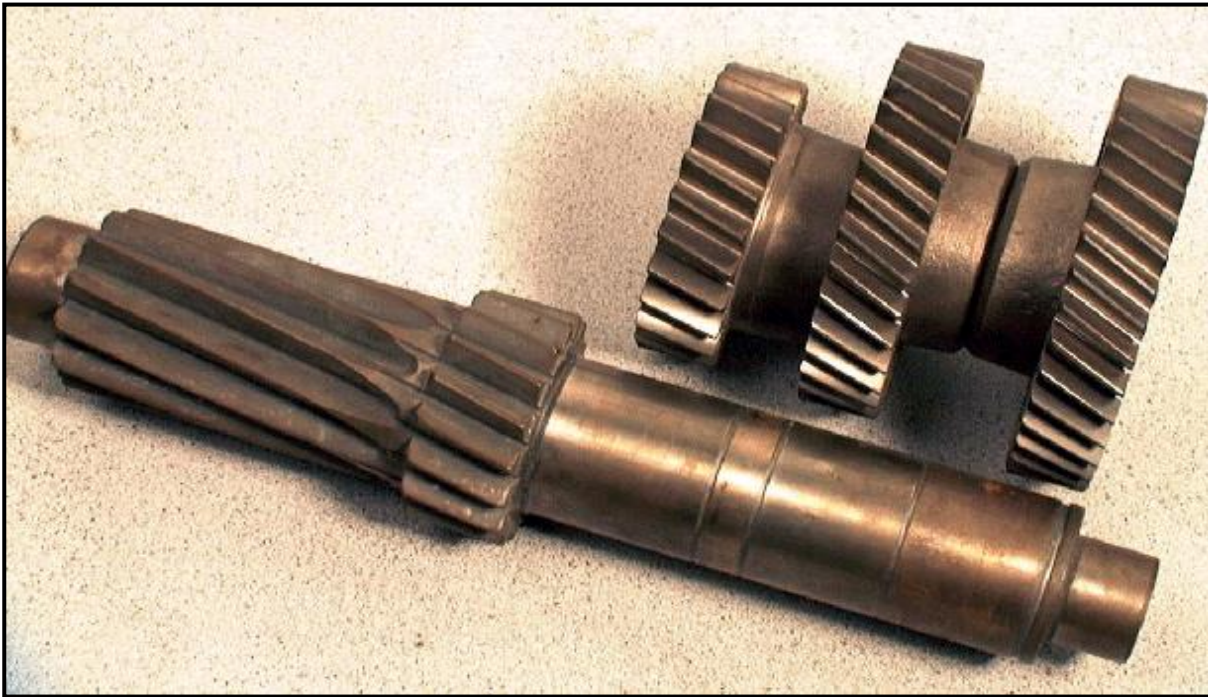


Rod and Tube Joint Design

Rod and tube joint design	Performance
	Good for rods in tension, compression, and tension if area is large; poor in bending
	Good for rods
	Good for rods
	Good for rods or tubes
	Poor for tubes
	Good for rods but slower to produce
	Good for tubes if wall thickness permits machining
	Corners of rods or tubes best handled with elbow

Cylindrical bonding

- Scania Truck Gearbox
- Press Bonded Gear Set onto Countershaft

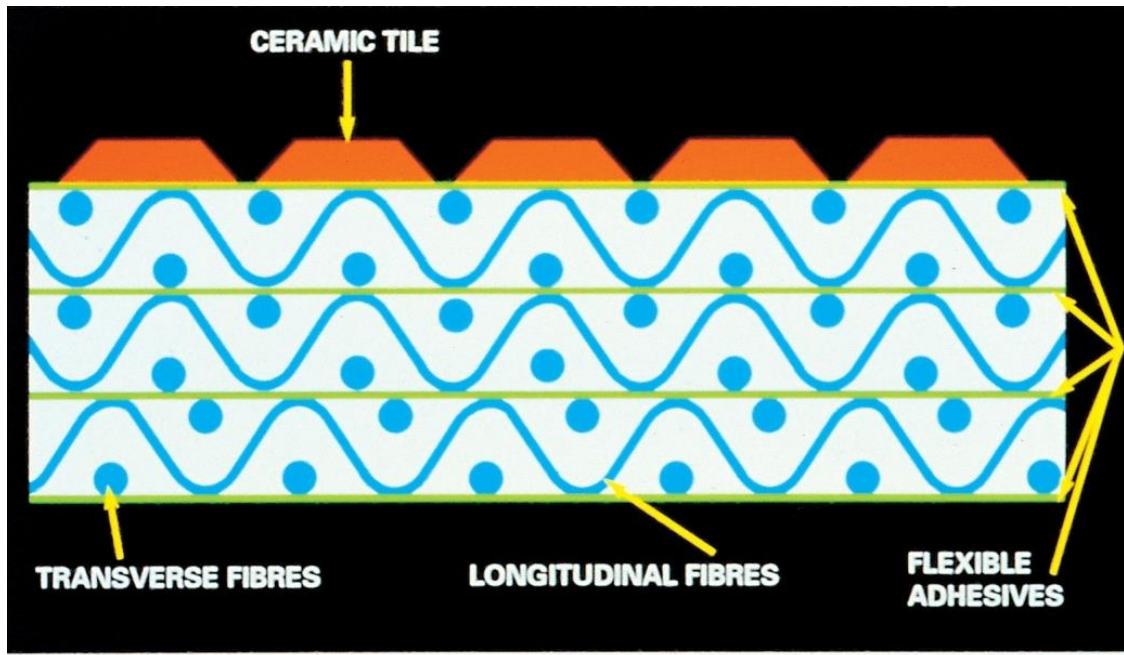


Threadsealing

- Threadsealing on petrol forecourt pump
- Safety critical



Bullet Proof Vest



Wind Turbine Blades

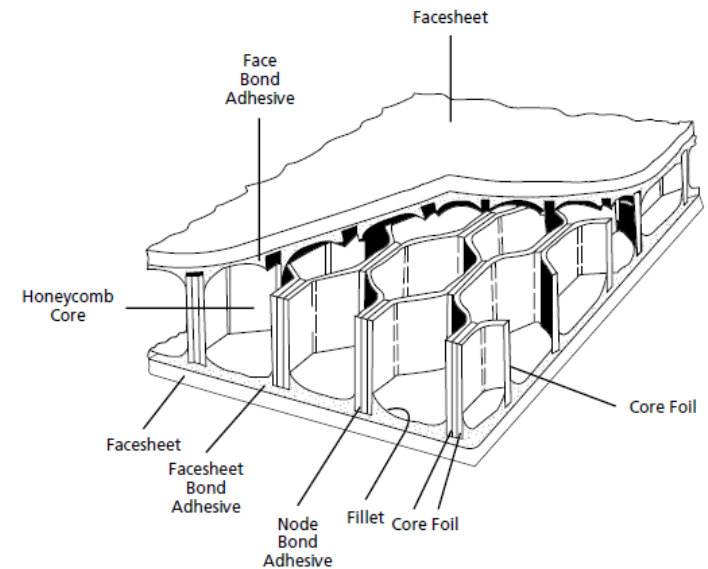
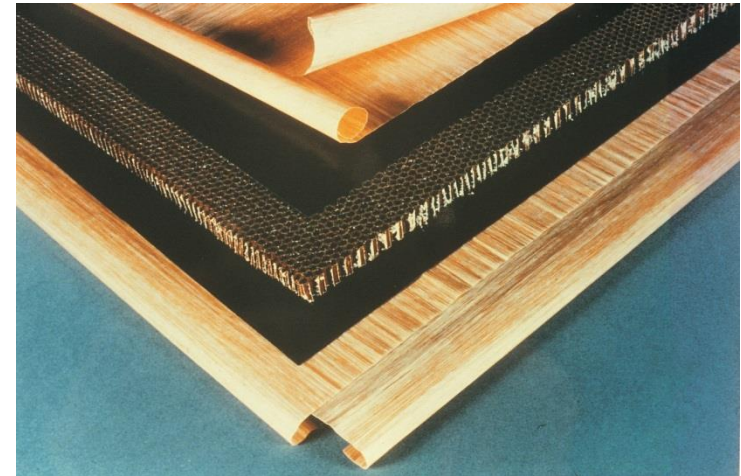


Cylindrical bonding of bicycle forks



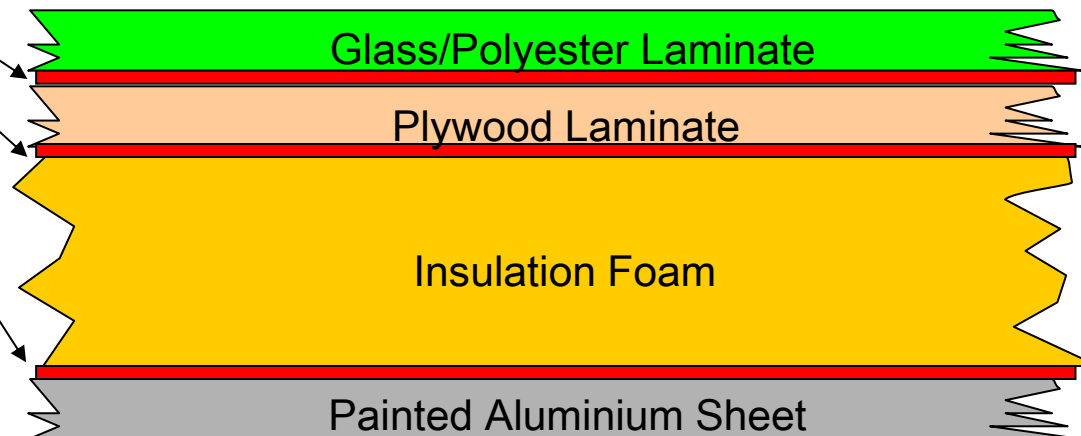
Aircraft Floor Panels

- Honeycomb Sandwich
- Aircraft Floor
- Interior Panels

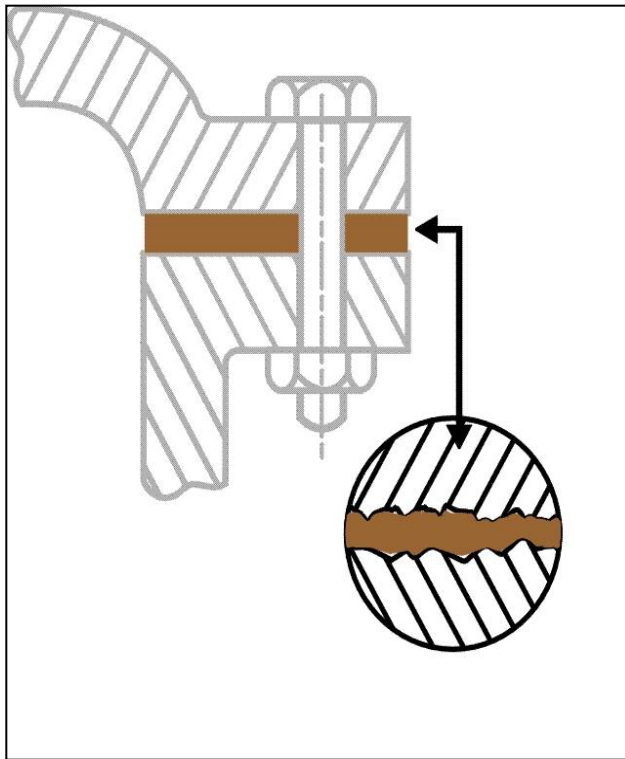


Structural Insulation Laminate

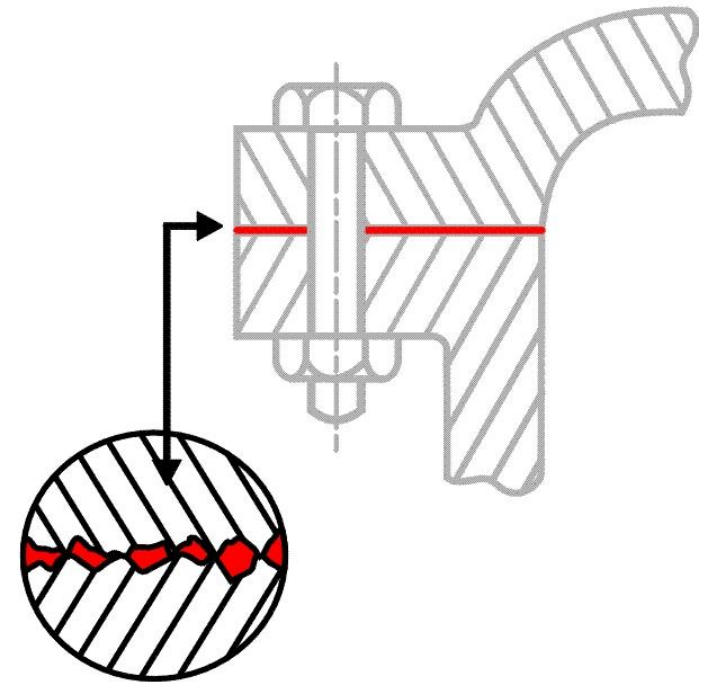
Adhesive
Bond
Lines



- How do they work?



Traditional



Liquid gaskets

UV Cure Adhesives in General Industry

- Optical Instruments
- Sports Equipment

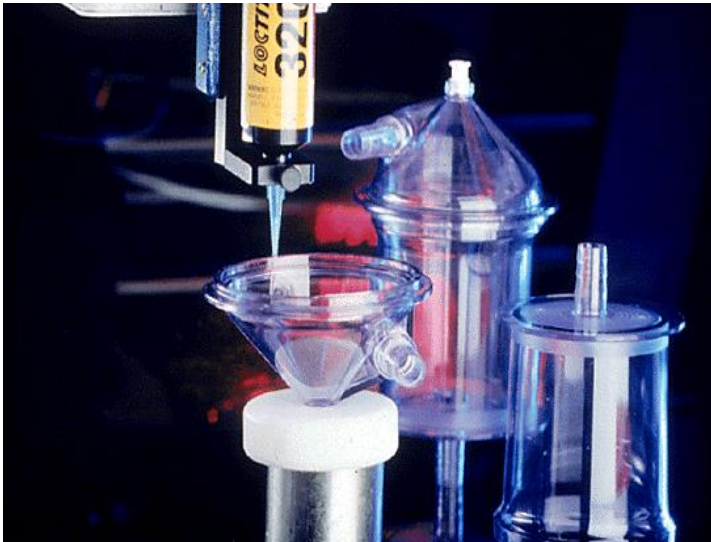


- Bonding & sealing of electrical connectors

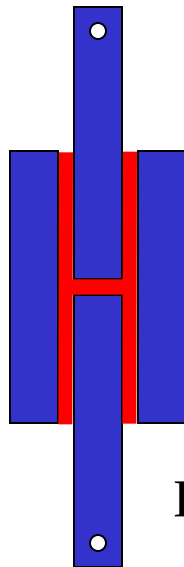


UV Cure Adhesives for the Medical Industry

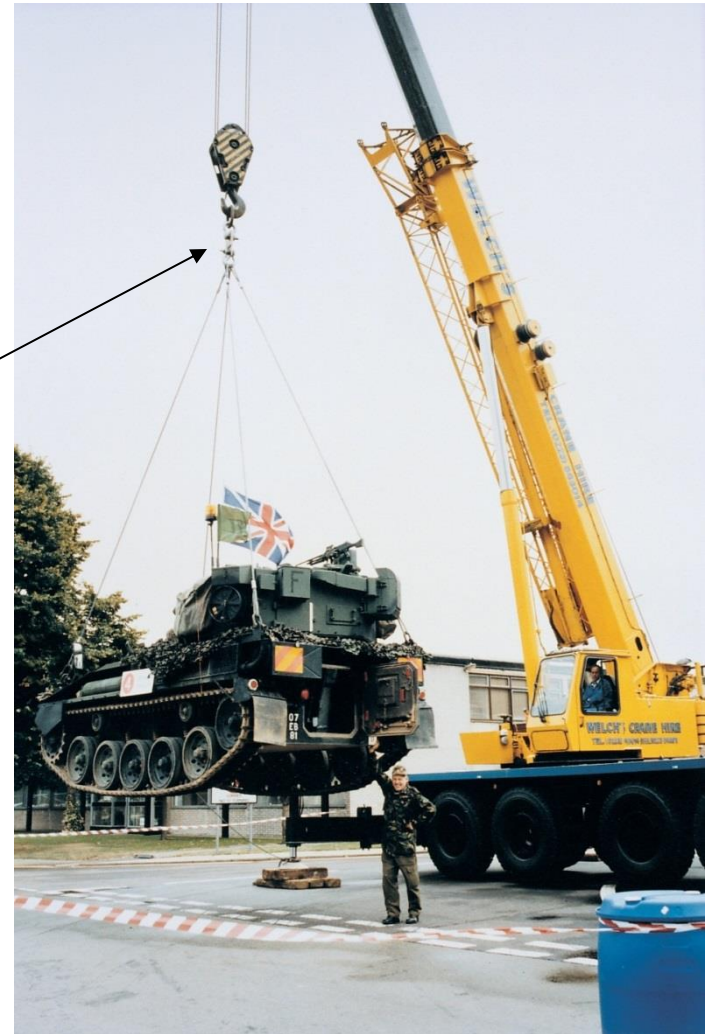
- Blood Filter (????)
- Face Mask (Silicone rubber) Bonding
- Needle Bonding



Double Lap Joint

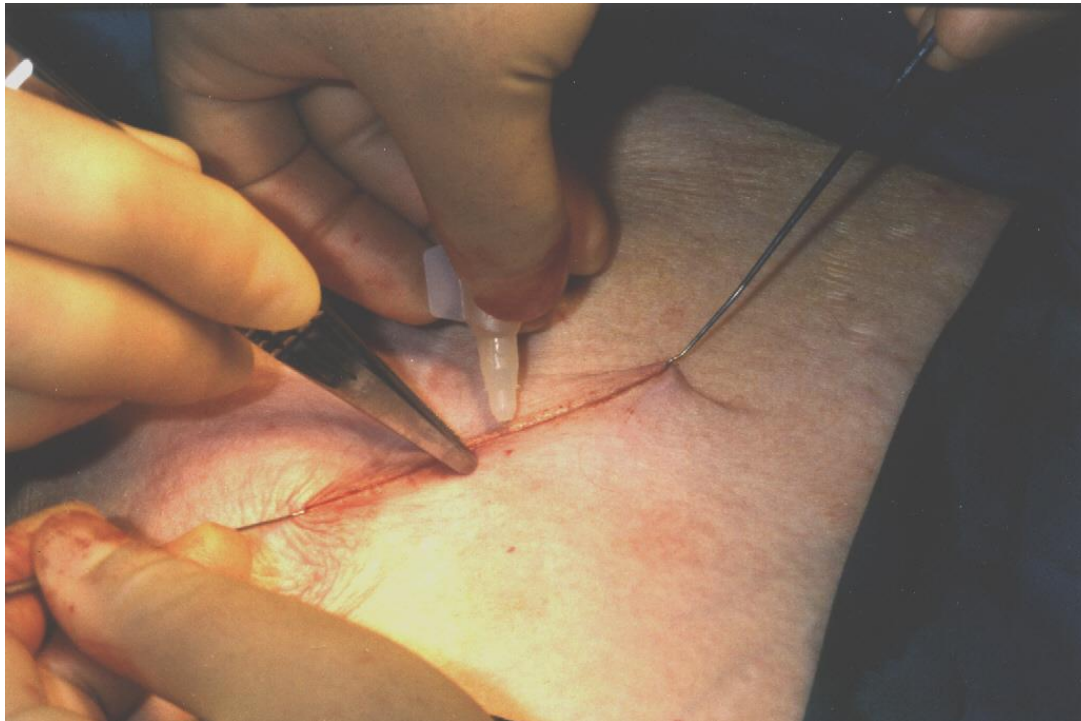


Double Lap Joint



Skin Bonding

- Adhesive Replaces Stitches
- Minimises Scarring
- Prevents Infection



Adhesives are used everywhere

